Use Case Pay Order By Credit Card

1. Use case code

UC004

2. Brief Description

This use case describes the interaction between the Customer, Paypal Sandbox System and AIMS System when the Customer wishes to pay for an order using a credit card.

3. Actors

3.1 Customer

3.2 Paypal Sandbox System

4. Preconditions

Customer in the Check Out page and choose the Credit Card Payment method

5. Basic Flow of Events

1. The software display invoice information

2. The Customer selects the "Credit Card" payment option and clicks "Proceed to Payment".

3. The software displays the credit card payment form.

4. The Customer enters their credit card details (see Table A) and clicks "Submit Payment".

5. The software validates the input data locally and sends a payment authorization request to the PayPal Sandbox API.

6. The PayPal Sandbox processes the simulated transaction.

7. The software notifies the payment result.

6. Alternative flows

Table N-Alternative flows of events for UC Place order

No	Location	Condition	Action	Resume location
1.	At Step 5	If the input data is incomplete or violates validation rules	▪ Software highlights the invalid fields.	Resumes at Step 3

		(e.g., invalid card length, expired date).	▪ Software displays a specific error message (e.g., "Invalid card number").	
2.	At Step 5	If the software fails to connect to the PayPal Sandbox API (timeout or server error).	▪ Software logs the system error. ▪ Software displays a "Gateway connection error. Please try again later." message.	Use case ends
3.	At Step 6	If the PayPal Sandbox API declines the test card (e.g., insufficient funds test card).	▪ Software logs the failed transaction attempt. ▪ Software displays a "Payment Declined" error message.	Resume at Step 2

7. Input data

Table A - Input data of Credit Card

No	Data fields	Description	Mandatory	Valid condition	Example
1.	Card Number	The 16-digit number on the front of the card.	Yes	Exactly 16 numeric digits. Must pass Luhn algorithm check.	4111 1111 1111 1111
2.	Expiration Date	The month and year the card expires.	Yes	Format MM/YY. Must be a future date.	03/30
3.	CVV	Card Verification Value (security code).	Yes	Exactly 3 or 4 numeric digits.	123

8. Output data

No	Data fields	Description	Display format	Example
1	Payment Status	Current status of the payment transaction	Text message (Success/Failure/Cancelled)	"Success"

9. Postconditions

The payment result is displayed successfully